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Mindfulness-based cognitive therapy for perceived stress and psychosomatic symptoms in Chinese adolescent girls: a mixed-methods school-based study of parent and teacher perceptions

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Abstract

Background Mindfulness-Based Cognitive Therapy (MBCT) integrates mindfulness practices with cognitive strategies to alleviate perceived stress and its associated symptoms. While well studied in adults, evidence for school-based MBCT among adolescents with psychosomatic complaints—particularly in China—remains limited.

Methods We conducted a quasi-experimental, mixed-methods study in three public secondary schools in DaZhou (Sichuan, China). Female students aged 14–16 were screened ($N = 1,200$) using the Psychosomatic Complaints Scale (PCS; cutoff > 45) and DSM-5–informed clinical interviews; 60 eligible students were randomly allocated to MBCT (8 weekly 75-minute sessions) or school-as-usual control. To mitigate non-specific effects, controls received attention-balanced administrative contacts (weekly neutral check-ins without psychological content). Outcome assessors and the data analyst were blinded to allocation. The primary outcomes were the Perceived Stress Scale (PSS; $\alpha = 0.85$) and the PCS ($\alpha = 0.86$), both of which were adapted for Chinese adolescents. Parent ($n = 6$) and teacher ($n = 5$) interviews were analyzed thematically (Braun & Clarke).

Results Compared with controls, the MBCT group showed greater reductions in perceived stress and psychosomatic symptoms at post-test and two-month follow-up (Group $\times$ Time, $p < .001$; partial $\eta^2 \approx 0.32$ – 0.63). Mean reductions exceeded a conventional 0.5 SD benchmark, supporting applied relevance alongside statistical significance. Qualitative analysis yielded six themes that contextualized change: (1) reduced stress reactivity, (2) improved emotion regulation and self-awareness, (3) strengthened coping, (4) enhanced peer/family communication, (5) better classroom engagement, and (6) conditions for maintenance (practice dose, prompts, family/teacher supports).

Conclusions School-delivered MBCT was associated with sustained improvements in perceived stress and psychosomatic complaints among 14–16-year-old girls, with convergent qualitative evidence from parents and teachers. Interpretation remains cautious given the passive control, short follow-up, and bounded generalizability.

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Future work should test MBCT against attention-matched/active comparators, extend follow-up to 6–12 months, and examine mechanisms (e.g., decentering, rumination) and cultural/implementation outcomes to inform scaling in diverse Chinese school contexts.

Keywords Adolescents, Psychosomatic symptoms, Perceived stress, Mindfulness-based cognitive therapy, School-based intervention, Mixed-methods, China

Introduction

In recent years, the mental health of students with chronic emotional and psychosomatic disorders has become a growing concern among educators, families, and policymakers [1]. Among these students, the interplay between psychological stress and physical symptoms such as fatigue, headaches, sleep disturbances, and gastrointestinal issues often complicates diagnosis and intervention [2]. These manifestations are commonly observed in adolescents with undiagnosed or under-supported mental disorders, who experience heightened perceived stress and emotional dysregulation in academic settings, which can further exacerbate their psychological and physical well-being [3]. *To maintain conceptual clarity and sampling precision, the present study focuses on Chinese secondary-school adolescents aged 14–16 years, and all references to the target group consistently use this age range.*

Perceived stress—a subjective evaluation of environmental demands exceeding one's coping resources—has been identified as a major contributing factor to poor mental health in adolescents [4]. This form of stress not only intensifies existing emotional problems but also contributes to somatization, blurring the line between physical and psychological distress [5]. Adolescents are particularly vulnerable to these effects due to ongoing developmental changes, making them more susceptible to stress-related disorders. *Because school is a primary context of daily stress for 14–16-year-olds, effective support should emphasize emotion regulation and self-management skills that are feasible within routine school schedules [6].*

Mindfulness-Based Cognitive Therapy (MBCT) has emerged as a promising approach for improving emotional regulation, reducing perceived stress, and enhancing mental health, particularly among adolescents with internalizing symptoms [7]. Originally developed to prevent relapse in depression, MBCT integrates cognitive therapy with mindfulness techniques to cultivate present-moment awareness and reduce automatic, maladaptive reactions to stress [8]. Recent trials indicate reductions in anxiety, depression, and perceived stress in student samples, alongside gains in attention, emotion regulation, and resilience [9, 10]. *However, evidence specific to school-delivered MBCT for Chinese adolescents aged 14–16 presenting with psychosomatic complaints remains limited, indicating a clear implementation gap in this setting.*

Students with psychosomatic and emotional disorders may struggle to articulate their distress, making adult observations valuable complements to self-report [11]. Teachers often detect behavioral changes in classrooms, while parents observe patterns at home [12]. *In this study, we incorporate concise input from parents and teachers to complement adolescent self-reports and to gauge whether any improvements generalize across school and home contexts [13].*

Although MBCT shows benefits for stress and emotional health, few investigations have systematically examined how such interventions are appraised by the adults who support adolescents beyond the therapeutic context [14, 15]. Furthermore, adolescents with psychosomatic symptoms are underrepresented in intervention research, despite indications that perceived stress mediates links between cognitive vulnerabilities and poor mental health outcomes [16, 17]. *Addressing this evidence gap can refine tools tailored to the needs of 14–16-year-old students with psychosomatic complaints in Chinese schools [18].*

Accordingly, this study evaluates a school-based MBCT program for Chinese adolescents aged 14–16 with psychosomatic symptoms, testing its effectiveness in reducing perceived stress and improving mental health. Outcomes are triangulated using adolescent self-reports and streamlined evaluations from parents and teachers to establish ecological validity within everyday school and home environments.

Literature review

Mindfulness-based cognitive therapy (MBCT) is commonly theorized as a dual-mechanism intervention that couples nonjudgmental, present-moment awareness with cognitive reappraisal to interrupt stress–affect–symptom cycles. This account is consistent with work on reversing maladaptive neuroplasticity in persistent mood disturbance [18] and with evidence that acceptance- and mindfulness-based self-help can reduce social anxiety symptoms when compared with inactive controls [19]. Neurodevelopmental findings further indicate that mindfulness practice modulates large-scale functional connectivity in children and adolescents—changes plausibly linked to improved regulatory capacity relevant to psychosomatic complaints [20]. Because change must generalize beyond the therapy room, adjunctive programming for key adults (parents and teachers) is often emphasized

to scaffold practice and reinforce coping in everyday settings [21].

Adult clinical trials underscore efficacy while surfacing significant design contrasts. For example, a randomized comparison of MBCT versus active stress-management training in obsessive–compulsive disorder supports disorder-specific benefits beyond nonspecific stress relief [22]. In contrast, correlational and experimental work have connected mindfulness with self-compassion and well-being, even among non-meditators, suggesting broad transdiagnostic relevance [23]. Large-scale syntheses demonstrate that mindfulness-based interventions (MBIs) confer benefits across various psychiatric conditions, although with notable heterogeneity by diagnosis, comparator, and setting [24–26]. Experimental paradigms demonstrate accelerated recovery from induced negative affect following mindfulness training [27], and studies of trauma-exposed individuals highlight the roles of avoidance and dispositional mindfulness in internalizing symptoms—mechanistic targets directly addressed in MBCT [28]. Comparisons of metacognitive therapy and mindfulness meditation reveal process differences (explicit belief modification versus decentering and attentional training), clarifying which therapeutic ingredients may be most important for somatic amplification [29]. Programmatic reflections on MBCT likewise call for clearer articulation of active components and implementation standards—concerns that inform adolescent adaptations [30].

Delivery format and context also shape outcomes. Tele-delivered MBCT shows feasibility and preliminary efficacy, illustrating how access barriers can be reduced without abandoning core elements [31]. Culturally diverse randomized trials (e.g., among depressed adults with disabilities in Nigeria) demonstrate that MBCT's benefits are not limited to WEIRD samples, reinforcing the need for local adaptation rather than wholesale reinvention [32]. At the same time, studies in young adults have linked daily stress, affect, and health behaviors, underscoring the importance of targeting stress-reactive routines alongside symptoms [33]. Meanwhile, gendered coping differences suggest potential moderation of intervention effects that developers should anticipate in school settings [34]. Population studies of suicide risk and eating-disorder symptomatology in college cohorts, together with evidence on high sedentary time and its mental-health correlates, further situate stress and health behaviors as intertwined targets—aspects often co-addressed by MBIs [35, 36]. Classic stress-process theory provides a framework for interpreting these multilevel influences and for designing multi-informant assessments that capture cross-context generalization [37].

Evidence specific to psychosomatic presentations strengthens the rationale for MBCT in youth with

somatic complaints. Randomized and clinical trials in irritable bowel syndrome have shown reduced symptom severity and improved quality of life following MBCT, with convergent theoretical implications for targeting interoceptive attention and catastrophic appraisal [38, 39]. Measurement choices matter when tracking co-occurring internalizing symptoms; validated tools used in primary care (e.g., PHQ-9) illustrate trade-offs between brevity and construct coverage that adolescent studies must weigh carefully [40]. Because coping resources and family support shape symptom expression and maintenance, contextual factors such as family functioning and stress burden are pertinent covariates or moderators in school-based trials [41]. Financial and role-strain stressors also intersect with mental health in students, highlighting the importance of assessing a broader stress ecology when interpreting the effects of MBCT [42]. Replicated gender patterns in coping advise attention to subgroup responsiveness and potential tailoring [34]. Sleep, a frequent co-occurring complaint in psychosomatic profiles, is a relevant secondary endpoint in adolescent studies given its bidirectional links with stress and mood [43]. Preventive mental health efforts in scholastic settings (e.g., eating disorder prevention) demonstrate that universal programs can be effective. Still, their effects depend on fit, fidelity, and the salience of outcomes for the target group [44]. Long-term maintenance observed in pain-related conditions (e.g., primary headache) suggests that when practice continues, benefits can persist—an argument for booster sessions and home–school practice bridges in youth protocols [45]. Contemporary psychosomatic psychiatry underscores mind–body bidirectionality, reinforcing integrated designs that track both psychological and somatic endpoints [46]. Within China, stressors such as bullying and cyberbullying are linked to poorer mental health in adolescents, contextualizing why school-embedded, developmentally sensitive MBCT may be warranted [47]; more broadly, research on populations facing major psychosocial stressors illustrates how context can complicate symptom interpretation and intervention transferability—cautionary notes when generalizing adult evidence to younger or culturally distinct groups [48].

Recent syntheses and adolescent-focused studies help refine adaptation priorities. Integrative and meta-analytic reviews call for sharper tests of mechanisms (e.g., decentering, rumination) and stress the need for age-appropriate dosing, active comparators, and longer follow-up in youth [49–51]. In China, a randomized classroom trial combining mindfulness with life skills training improved emotion regulation and reduced anxiety in secondary students, demonstrating the feasibility of this approach in real-time schedules and highlighting the value of layered skills alongside mindfulness practice

[52]. Complementary school-delivered work in middle-schoolers reported reductions in learning burnout, hinting at academic relevance when protocols align with curricular rhythms [53]. Brief digital adaptations in Chinese adolescents have demonstrated short-term prosocial benefits, offering a scalable adjunct to in-person sessions. Meanwhile, meta-analytic evidence in Chinese college students supports anxiety reduction, but cautions against direct generalization to younger adolescents without developmental tailoring of framing, practice, and support [54, 55]. Extensive cluster-randomized evaluations of universal school mindfulness (e.g., MYRIAD; ~8,400 students) report no advantage over teaching-as-usual at one year, highlighting how instructor training, delivery mandate, and home practice can mute effects [56]. Other cluster designs with stronger fidelity support yield small but positive effects, and umbrella reviews emphasize cultural/contextual adaptation, as well as implementation quality—key considerations for academically intensive Chinese secondary schools and for adolescents with psychosomatic complaints whose symptoms fluctuate with stress exposure [57, 58].

Addressing the methodological concerns raised in prior work, our design (i) targets a clearly defined developmental window (ages 14–16), (ii) uses a school-delivered MBCT protocol with instructor training and fidelity safeguards, (iii) specifies psychosomatic and stress outcomes a priori, (iv) incorporates multi-informant assessment (student, teacher, parent) to capture home–school generalization, and (v) situates interpretation within contextual stress ecology [59, 60]. These choices are intended to move beyond cataloging efficacy toward a synthesis that tests theoretically specified mechanisms while attending to the cultural and developmental realities of Chinese secondary schools.

Objectives

Guided by a convergent mixed-methods design, this study pursues three interlocking objectives. First, it will determine whether a school-delivered MBCT program produces greater reductions than a control condition in perceived stress and psychosomatic symptom burden among secondary-school adolescents aged 14–16, with “mental health” explicitly operationalized as Perceived Stress Scale (PSS) total and Psychosomatic Complaints Scale (PCS) total/subscales measured at baseline, post-intervention, and follow-up. Second, in parallel, it will qualitatively examine how students, parents, and teachers perceive changes in emotion regulation, coping, classroom functioning, and somatic complaints, while also appraising the acceptability and feasibility of MBCT through semi-structured interviews. Finally, to move beyond parallel findings toward an integrated account, the study will synthesize quantitative and qualitative

results—using joint displays and triangulation—to evaluate the convergence between measured improvements (PSS/PCS) and stakeholder-reported changes, thereby interpreting outcomes through the study’s theoretical lens of attention regulation, decentering, and cognitive reappraisal within the everyday school ecology. Guided by the study aims, we state the following hypotheses:

H1 MBCT produces a significant Group (MBCT vs. control) $\times$ Time (baseline, post-test, follow-up) interaction on the Perceived Stress Scale (PSS) and the Psychosomatic Complaints Scale (PCS), with greater reductions in the MBCT group.

H2 After adjustment for baseline (and prespecified covariates), the MBCT group shows lower PSS and PCS means than the control group at post-test and follow-up.

H3 A higher proportion of MBCT participants achieves clinically meaningful improvement on PSS and PCS (e.g., ≥ 0.5 SD reduction or reliable change) than controls.

H4 =parents and teachers report improvements in students’ emotion regulation, coping, classroom functioning, and management of somatic complaints after MBCT relative to control.

Method

Sampling and design

This study adopted a mixed-methods approach, integrating a quasi-experimental quantitative design with a complementary qualitative component to examine the effectiveness and contextual perceptions of Mindfulness-Based Cognitive Therapy (MBCT) for students with psychosomatic disorders. The research was conducted in DaZhou, SiChuan Province, during the 2022–2023 academic year and targeted female students in the second stage of secondary education. *To eliminate prior inconsistencies and protect internal validity, the target age range is standardized to 14–16 years throughout the study.* A two-phase selection process was employed: first, stratified random sampling recruited 1,200 students from multiple public secondary schools; second, purposive selection identified those who met the inclusion criteria.

In the quantitative phase, a pretest–posttest control-group design with a two-month follow-up was employed. The Psychosomatic Complaints Scale (PCS) was administered to all 1,200 students; those scoring above the clinical threshold (score > 45) were shortlisted. From this pool, 70 students completed clinical interviews with trained school psychologists in line with DSM-5 criteria; 60 students were eligible and randomly assigned to the experimental ($n = 30$) or control group ($n = 30$) via computerized randomization executed by a researcher independent of

data collection. *Allocation concealment used sequentially numbered, opaque, sealed envelopes (SNOSE) prepared by an independent coordinator. Because participants necessarily know whether they attend sessions, participant blinding to the condition is not feasible. Instead, we blind them to the study hypotheses and the expected direction of effects. We also blind outcome assessors and the data analyst to group allocation to minimize detection bias.*

An a priori power analysis (G*Power 3.1) indicated that 26 participants per group provide 0.80 power to detect a medium effect ($f = 0.25$) at $\alpha = 0.05$; thus, the randomized sample is adequate. The experimental group received MBCT in eight weekly 75-minute sessions delivered by a licensed mindfulness practitioner; fidelity was monitored via a standardized MBCT adherence checklist. The control group continued their usual school schedule without added programming. *Because the control condition is passive, we explicitly acknowledge the risks of expectancy and attention (placebo) effects. To attenuate these, assessment contact, scheduling communications, and incentives were standardized across groups; however, non-specific effects cannot be entirely excluded and are treated as a design limitation a priori.* Three participants withdrew (two experimental, one control), leaving 57 completers (28 experimental, 29 control).

Eligibility required: PCS > 45; no current psychiatric medication/psychotherapy; age 14–16 years; student assent and guardian consent; and commitment to session attendance. Exclusion criteria were > 3 absences, disqualifying medical conditions, or refusal to complete follow-up. Group means did not differ at baseline on key measures.

To enrich interpretation, a qualitative case-study strand was purposively sampled, comprising six parents (three mothers, three fathers) of experimental-group students and five teachers (two counselors, three subject teachers). Semi-structured interviews were conducted ~1 week after the intervention, lasting 30–45 min, and took place in private rooms at the school. The guide elicited observations on behavioral, emotional, and academic change; implementation challenges; and recommendations for school deployment. *Data adequacy is defined a priori as thematic saturation (no new codes/themes across two consecutive interviews); saturation was reached within the 11 interviews, and an audit trail was maintained to support dependability.*

Instruments

To gather robust data across both quantitative and qualitative phases, this study employed two standardized self-report questionnaires and a semi-structured interview protocol. The instruments were selected for their conceptual relevance, previous psychometric validation, and suitability for use with Chinese adolescent populations.

Where necessary, cultural and linguistic adaptations were made to enhance contextual appropriateness, and all tools underwent preliminary pilot testing to assess clarity, relevance, and cultural fit.

Psychosomatic complaints scale (PCS)

The Psychosomatic Complaints Scale (PCS), developed by Takada and Sakata [61], is designed to assess psychosomatic symptoms in adolescents. It consists of 30 items, each scored on a 4-point Likert scale, ranging from “never” (0) to “repeatedly” (3), with total scores ranging from 0 to 90. A cutoff score of 45 indicates potential psychosomatic dysfunction. Initial studies reported a Cronbach’s alpha of 0.64 and confirmed a unifactorial structure with one factor explaining 34.1% of the variance [61].

Subsequent standardization of the scale by Fruehwirth [62] confirmed its content validity through expert panel review and concurrent validity with the Goldberg General Health Scale. Hajloo [63] conducted exploratory and confirmatory factor analyses in an Iranian context, confirming its one-dimensionality (KMO = 0.782) and high reliability ($\alpha = 0.85$; $r = .85$ for test–retest reliability). In the present study, the PCS underwent further psychometric evaluation, yielding a Cronbach’s alpha of 0.86 and acceptable model fit indices in confirmatory factor analysis (CFI = 0.93, RMSEA = 0.06), thus empirically supporting its reliability and validity for the Chinese adolescent sample. *Before the main data collection, the PCS was forward- and back-translated by bilingual experts and reviewed by two Chinese clinical psychologists to ensure cultural and linguistic appropriateness.* To strengthen face validity, beyond the original 10-student pilot, the adapted version was tested with 25 non-participating students representative of the main sample, leading to the rephrasing of two ambiguous items.

Perceived stress scale (PSS)

The Perceived Stress Scale (PSS), developed by Cohen et al. [64], consists of 14 items, each rated on a 5-point scale (0 = “never” to 4 = “very often”), yielding a total score ranging from 0 to 56. Higher scores reflect greater perceived stress. Internal consistency in the original development sample ranged from 0.84 to 0.86 [65]. Broton et al. [42] validated the scale cross-culturally, achieving acceptable reliability ($\alpha = 0.81$ in the Iranian sample and 0.86 in the American sample).

In the present study, the PSS was adapted using the same rigorous forward–backward translation procedure applied to the PCS. Its content validity was reviewed by a panel of local mental health experts with experience in assessing adolescent stress. Cognitive interviews were conducted with 15 students to evaluate the semantic clarity, cultural appropriateness, and comprehension of the

items. The Cronbach's alpha in this study was 0.85, confirming satisfactory internal consistency. A confirmatory factor analysis further validated the two-factor structure of the PSS (CFI = 0.92, RMSEA = 0.05).

Semi-structured interview protocol

To support the qualitative aims, a semi-structured interview protocol was developed based on themes drawn from MBCT literature, school-based intervention studies, and adolescent psychosocial frameworks. The guide included 10 open-ended items addressing emotional, behavioral, and social changes, perceived effectiveness of MBCT, observed challenges, and contextual fit in school settings.

To establish content validity, the protocol was reviewed by two qualitative research specialists and two school psychologists with experience in adolescent mental health. Pilot testing involved one parent and one teacher, resulting in minor revisions to improve the clarity and sensitivity of the wording. While no psychometric indices apply to open-ended protocols, *trustworthiness was enhanced through expert validation, member checking after transcription, and adherence to Braun and Clarke's thematic analysis procedure* [66]. The final interview protocol was designed to elicit nuanced reflections on both individual and contextual factors affecting the MBCT experience (See appendix).

Intervention

We delivered an eight-session MBCT program (75 min per week) aligned with the standard manual and adapted for secondary school adolescents. Each session integrated a core mindfulness practice (e.g., breath/body awareness, mindful eating), targeted cognitive exercises (working with distressing thoughts and emotions, decentering), and a brief group discussion to consolidate learning. Licensed mental-health professionals with formal MBCT training facilitated all sessions. We verified facilitator credentials before delivery, provided biweekly supervision throughout the trial, and monitored adherence using an MBCT fidelity checklist, structured observation notes, and facilitator logs.

Home practice was integral. Students received daily exercises and a brief reflective journal. We tracked adherence through weekly self-reports and facilitator notes. When practice fell below four of seven days in a week, facilitators conducted a ≤ 10 -minute motivational check-in to identify barriers, offered 3–5-minute “micro-practices,” and set a concrete goal for the following week; repeated shortfall over two consecutive weeks prompted tailored adjustments (shorter practices, alternate times of day). All deviations and resolutions were logged and later treated as fidelity covariates in sensitivity analyses.

The control condition was school-as-usual (passive comparator). Because passive controls can introduce non-specific influences (expectancy/attention effects), we implemented attention-balancing procedures a priori: both groups received identical administrative contact (weekly scheduling and well-being check-ins, ≤ 10 min with no psychological content), all study communications used neutral language, and the control group was offered delayed MBCT after the trial. To minimize contamination, intervention and control activities occurred in separate rooms and time slots, and staff not involved in delivery were instructed not to discuss session content with students. We acknowledge that non-specific effects cannot be fully eliminated without an attention-matched or active control; consequently, we have pre-specified these limitations in the Methods section and planned an active, education-based comparator for future iterations.

Procedure

We obtained approvals from the DaZhou (SiChuan) education authority and the university ethics board (Approval No. 2022-EDU-MBCT-18) and coordinated with three public girls' secondary schools. Screening proceeded in two steps: first, the Psychosomatic Complaints Scale (PCS) was administered to 1,200 students; second, those scoring > 45 completed DSM-5-based clinical interviews with trained school psychologists. Sixty eligible adolescents (aged 14–16) were randomized (30 MBCT; 30 control) using a computerized sequence prepared by an independent coordinator. Allocation concealment was used sequentially numbered, opaque, sealed envelopes. Because students necessarily knew whether they attended sessions, we blinded the outcome assessors and the data analyst to allocation and provided participants with neutral study aims to limit expectancy effects.

The timeline comprised (1) baseline assessment, (2) the eight-week intervention, and (3) post-test plus a two-month follow-up. Sessions ran after school in quiet, ventilated classrooms. We maintained fidelity logs and selective audio recordings to verify adherence to the manual. The control group followed the routine timetable without added programming; both groups received the matched administrative contact described above. Attrition was low (three withdrawals), and an attrition analysis showed no baseline differences between completers and non-completers.

All procedures were free of charge. We obtained written assent from students and consent from guardians, assigned anonymized IDs, and reported only aggregated data. COVID-19 precautions (distancing, masking, ventilation, and sanitization) were applied uniformly across groups; no sessions were cancelled or moved online, and

there was no differential attrition or missingness attributable to these measures.

For quantitative outcomes, we used mixed ANOVA (Group $\times$ Time) with assumption checks and Bonferroni-adjusted pairwise comparisons, reporting partial η^2 and confidence intervals. To strengthen missing-data handling beyond intention-to-treat and last-observation-carried-forward, we employed multiple imputation as the primary strategy ($m=20$; wholly conditional specification under MAR) and reported LOCF and complete-case analyses as sensitivity checks; planned covariates (baseline PSS/PCS, age, school) were included in adjusted models. For qualitative data, we conducted semi-structured interviews with parents and teachers ~ 1 week post-intervention. We analyzed the transcripts in MAXQDA, following Braun and Clarke's six-phase approach, with dual independent coding and negotiated agreement. We used member checking with a subset of participants and documented thematic saturation and decision trails in an audit log. The content of mindfulness-based cognitive therapy is presented in Table 1.

In addition, after completing the quantitative phase, semi-structured interviews were conducted with 11 selected informants, comprising 6 parents (3 mothers and 3 fathers) and 5 teachers (2 school counselors and 3 subject-area teachers). The purpose was to gather qualitative data on their perceptions of MBCT. *The selection of the informants was based on their direct involvement with the students in the experimental group, ensuring a range of perspectives from both home and school settings.* The interviews were designed to explore their observations, experiences, and opinions regarding the effects of MBCT on students' perceived stress levels and psychological symptoms. *Interviews were conducted in a private room at the school, lasting between 30 and 45 min. They were audio-recorded with the participants' informed consent.*

The interview process allowed participants to share their perspectives openly and provide detailed accounts

of their observations. *To ensure rigor in the qualitative analysis, a team of two independent researchers conducted the coding process using Braun and Clarke's thematic analysis framework. Both coders were blinded to the participants' group status (experimental or control) and the study's hypotheses to minimize potential bias from the interviewer.* Triangulation was employed by comparing the perspectives of parents, teachers, and the students' quantitative outcomes to enhance the trustworthiness of the findings. *Data saturation was reached after the 11 interviews, as no new themes emerged during the analysis, indicating that sufficient data had been collected to represent the range of participant experiences.*

Potential interviewer bias and reflexivity were addressed through regular team discussions, where coders reflected on their positionality and ensured that their preconceptions did not influence the coding process.

Data analysis

We analyzed quantitative and qualitative data using procedures aligned with the study aims and best practices for credibility and transparency.

Quantitative phase

We conducted analyses in SPSS 26.0 (two-tailed $\alpha = 0.05$; 95% CIs). After descriptive summaries (means, SDs, frequencies), we screened assumptions and data quality: visual outlier checks (boxplots; $|z| > 3$ flagged for review), Shapiro–Wilk for normality, Levene for homogeneity, and Mauchly for sphericity (with Greenhouse–Geisser corrections where violated). Primary tests used a mixed 2 (Group: MBCT, control) $\times$ 3 (Time: pre, post, follow-up) ANOVA to estimate main and interaction effects. Where interactions were significant, we ran Bonferroni-adjusted pairwise contrasts; we reported partial η^2 for omnibus effects and Hedges g with 95% CIs for pairwise comparisons.

Table 1 Summary of Mindfulness-Based cognitive therapy sessions

Sessions	Session Summary
First	Introduction and building trust, mindfulness as a way of life, eating exercises, feedback, and discussion on eating exercises.
Second	Discussion on the consistency of mindful activities, introduction to home exercises, discussion on mindfulness attitudes (non-striving, beginner's mind, etc.), introduction to seated Meditation, feedback, and discussion on meditation.
Third	Seated meditation, seeing and hearing exercises, discussion on mindfulness attitudes (non-judgment and acceptance), cultivating mindfulness: introduction of mindful dialogue as an exercise, feedback, and discussion.
Forth	Seated meditation: awareness of breath, body, sounds, and thoughts, discussion on mindfulness attitudes (acceptance and letting go), dealing with distressing emotions and feelings, feedback and discussion on acceptance and letting go.
Fifth	Seated meditation: awareness of breath, body, sounds, and thoughts, introduction of distressing thoughts and patterns, breathing exercises, feedback, and discussion on distressing thoughts.
Sixth	Seated meditation: mindfulness of thoughts, breathing exercises for anger management, mindfulness and emotions, contemplation, feedback, and discussion on anger Management.
Seventh	Seated meditation: awareness of breath, body, sounds, and thoughts, mindfulness and compassion, contemplation, feedback, and discussion on compassion.
Eighth	Seated meditation: conclusion, discussion on how to integrate mindfulness practices into one's lifestyle seamlessly

Because baseline severity and background factors can influence change, we ran adjusted models that controlled for baseline PSS/PCS, age, and school (fixed effect). In addition, to test robustness beyond repeated-measures ANOVA, we fit linear mixed-effects models (random intercept for participant; fixed Group, Time, Group×Time; same covariates) via SPSS MIXED. Results from mixed models converged with ANOVA in sign and inference; we report both in the Results (primary = ANOVA; mixed models = sensitivity).

Missingness was low (<5%) and passed Little’s MCAR test; nevertheless, given known biases of LOCF, we used multiple imputation as our primary strategy: $m = 20$ datasets via fully conditional specification (predictive mean matching for continuous variables), including Group, Time, outcomes at all waves, and covariates in the imputation model. Estimates were pooled using Rubin’s rules. We performed LOCF and complete-case analyses as sensitivity checks; the conclusions remained unchanged across both approaches. Where assumptions were materially violated, we confirmed patterns using heteroscedasticity-robust CIs and rank-based alternatives; these checks did not alter substantive interpretations.

Qualitative phase

Interview audio was transcribed verbatim and imported into MAXQDA 2022. We used Braun and Clarke’s six-phase thematic analysis with a combined deductive–inductive codebook: deductive nodes reflected the study lens (attention regulation, decentering, cognitive reappraisal; classroom and home transfer), while inductive nodes captured participant-generated concepts. Two analysts independently coded an initial 20% of transcripts, refined code definitions through memoing, and proceeded only after achieving Cohen’s $\kappa \geq 0.75$ on overlapping segments. The remaining transcripts were double-coded in rotating pairs; disagreements were resolved via negotiated agreement in weekly meetings, with decisions logged in an audit trail (versioned codebook, decision memos, and rationale).

We used MAXQDA’s Code Matrix Browser and Summary Grid to compare patterns across informant groups (parents vs. teachers) and to support triangulation with quantitative change (theme–outcome matrices linking interview excerpts to participants’ PSS/PCS trajectories). Member checking (with a subset of participants) and

peer debriefing (with a methodologist not on the coding team) enhanced credibility; reflexivity memos documented the analysts’ positionality and decisions.

Saturation was operationalized a priori as the point at which no new codes or subthemes emerged in two consecutive interviews. We tracked this with a saturation grid; saturation was reached within the 11 interviews, supporting sample adequacy for our focused aims. The final themes prioritized semantic coherence and relevance to the research questions (emotion regulation, coping, classroom functioning, and somatic complaint management), and were integrated with the quantitative results in joint displays to evaluate convergence and illuminate mechanisms.

Quantitative results

The means and standard deviations of the dependent variables—perceived stress and mental health—at pre-test, post-test, and follow-up for the experimental and control groups are presented in Table 2.

Table 2 presents the means and standard deviations of perceived stress and mental health scores at pre-test, post-test, and follow-up for both groups. In addition to these descriptive statistics, 95% confidence intervals (CIs) were calculated to indicate the precision of the observed effects. For perceived stress, students in the experimental group reported a significant reduction from pre-test ($M = 35.33$, $SD = 6.12$, 95% CI [33.01, 37.65]) to post-test ($M = 28.55$, $SD = 6.58$, 95% CI [26.06, 31.04]). Although scores slightly increased at follow-up ($M = 29.50$, $SD = 6.46$, 95% CI [27.06, 31.94]), they remained well below baseline levels. By contrast, the control group showed no meaningful change across time points (pre-test: $M = 33.42$, $SD = 4.79$, 95% CI [31.77, 35.07]; post-test: $M = 34.36$, $SD = 4.49$, 95% CI [32.82, 35.90]; follow-up: $M = 34.05$, $SD = 4.58$, 95% CI [32.49, 35.61]). *Using a distribution-based benchmark of ≥ 0.5 SD for clinical importance (≈ 3.06 on PSS), the experimental group’s mean reductions from pre-test to post-test ($\Delta = 6.78$) and to follow-up ($\Delta = 5.83$) exceeded this threshold, suggesting potential clinical relevance beyond statistical significance.*

A similar pattern emerged for mental health (where lower scores reflect fewer symptoms). The experimental group improved from pre-test ($M = 55.83$, $SD = 7.94$, 95% CI [52.71, 58.95]) to post-test ($M = 49.66$, $SD = 7.53$, 95% CI [46.70, 52.62]) and maintained this gain at follow-up

Table 2 Means and standard deviations of perceived stress and mental health across the three time points in experimental and control groups

Variable	Group	Pre-test (M ± SD)	Post-test (M ± SD)	Follow-up (M ± SD)
Perceived Stress	Experimental	35.33 ± 6.12	28.55 ± 6.58	29.50 ± 6.46
	Control	33.42 ± 4.79	34.36 ± 4.49	34.05 ± 4.58
Mental Health	Experimental	55.83 ± 7.94	49.66 ± 7.53	49.94 ± 7.58
	Control	57.21 ± 6.55	57.84 ± 6.67	57.94 ± 6.85

Table 3 Mixed ANOVA results for perceived stress and mental health across time and group

Variable	Source	SS	df	MS	F	p	η^2 (Effect Size)
Perceived Stress	Time	189.32	2	94.66	35.75	< 0.0001	0.50
	Group	220.16	1	220.16	28.42	< 0.0001	0.48
	Time $\times$ Group	317.54	2	158.77	59.96	< 0.0001	0.63
Mental Health	Time	176.63	2	88.31	20.38	< 0.0001	0.37
	Group	949.58	1	949.58	16.47	< 0.0001	0.32
	Time $\times$ Group	277.74	2	138.87	32.05	< 0.0001	0.48

Table 4 Bonferroni pairwise comparisons between time points for perceived stress and mental health

Variable	Comparison	Mean Difference	p-value
Perceived Stress	Pre-test vs. Post-test	2.91	0.001
	Pre-test vs. Follow-up	2.60	0.001
	Post-test vs. Follow-up	-0.31	0.21
Mental Health	Pre-test vs. Post-test	2.76	0.001
	Pre-test vs. Follow-up	2.57	0.001
	Post-test vs. Follow-up	-0.19	0.75

($M=49.94$, $SD=7.58$, 95% CI [46.97, 52.91]). In contrast, the control group's mental health scores remained virtually unchanged (pre-test: $M=57.21$, $SD=6.55$, 95% CI [54.93, 59.49]; post-test: $M=57.84$, $SD=6.67$, 95% CI [55.53, 60.15]; follow-up: $M=57.94$, $SD=6.85$, 95% CI [55.58, 60.30]). Applying the same ≥ 0.5 SD benchmark (≈ 3.97 on this scale), the experimental group's mean reductions (pre $\rightarrow$ post $\Delta=6.17$; pre $\rightarrow$ follow-up $\Delta=5.89$) met or exceeded the criterion, supporting applied interpretability of the changes.

To test for statistical significance, a mixed-design ANOVA (2×3) was conducted (Table 3). For perceived stress, there was a significant main effect of time, $F(2, 112)=35.75$, $p<.0001$, partial $\eta^2 = 0.50$, 95% CI [0.36, 0.60]; a significant main effect of group, $F(1, 56)=28.42$, $p<.0001$, partial $\eta^2 = 0.48$, 95% CI [0.31, 0.58]; and a highly significant time $\times$ group interaction, $F(2, 112)=59.96$, $p<.0001$, partial $\eta^2 = 0.63$, 95% CI [0.51, 0.70]. For mental health, significant effects were also found for time, $F(2, 112)=20.38$, $p<.0001$, partial $\eta^2 = 0.37$, 95% CI [0.22, 0.49]; for group, $F(1, 56)=16.47$, $p<.0001$, partial $\eta^2 = 0.32$, 95% CI [0.16, 0.45]; and for the time $\times$ group interaction, $F(2, 112)=32.05$, $p<.0001$, partial $\eta^2 = 0.48$, 95% CI [0.33, 0.58]. Adjusted mixed-effects models that controlled for baseline PSS/PCS, age, and school replicated all inferences, and multiple-imputation-pooled estimates converged with complete-case and LOCF sensitivity analyses (see Appendix), strengthening robustness.

Bonferroni-adjusted pairwise comparisons (Table 4) confirmed that perceived stress significantly decreased between pre-test and post-test (Mean Difference = 2.91, $p=.001$, 95% CI [1.88, 3.94]) and between pre-test and follow-up (Mean Difference = 2.60, $p=.001$, 95% CI [1.55, 3.65]); the small increase between post-test and follow-up (Mean Difference = -0.31, $p=.21$) was not significant,

indicating sustained stress reduction. For mental health, significant improvements were found from pre-test to post-test (Mean Difference = 2.76, $p=.001$, 95% CI [1.72, 3.80]) and from pre-test to follow-up (Mean Difference = 2.57, $p=.001$, 95% CI [1.48, 3.66]), with no significant change between post-test and follow-up (Mean Difference = -0.19, $p=.75$). Within-group Bonferroni tests indicated that time contrasts were significant for the experimental group but not for the control group, adding precision to the interpretation that change was intervention-linked rather than due to measurement fluctuation.

These findings collectively confirm that MBCT led to significant and sustained improvements in both perceived stress and mental health among students with psychosomatic symptoms. The lack of substantial differences between post-test and follow-up scores supports the durability of the intervention's effects over two months. To aid applied interpretation, we note that average reductions exceeded a half-standard deviation on both outcomes, a commonly used distribution-based marker of clinically meaningful change. Although the control group was broadly stable, this stability could reflect natural fluctuations around the baseline or modest expectancy/attention effects typical of passive comparators. We mitigated these risks through matched administrative contact; however, an attention-matched control would allow for a stronger test in future work. Notably, no participant attrition occurred between post-test and follow-up, ensuring consistency in sample size and enhancing the reliability of longitudinal comparisons.

Qualitative findings

Thematic analysis of interviews with 11 informants (6 parents and 5 teachers) yielded six overarching themes, following Braun and Clarke's six-phase model and supported by MAXQDA 2022. Open coding was conducted

independently by two researchers, followed by axial coding to develop coherent themes. To ensure the trustworthiness of the findings, triangulation across parent and teacher accounts was employed, and any deviant cases were addressed where relevant.

The participants were selected based on their direct involvement with the students in the experimental group, ensuring a range of perspectives from both home and school settings. The informants included a diverse sample of parents and teachers, representing both genders, a variety of professional roles, and a wide range of years of experience, enhancing the representativeness of the sample. Interviews were conducted in private settings, lasting between 30 and 45 min, and were audio-recorded with the informed consent of all participants.

Each theme is presented below with a concise explanation, illustrative participant quotations, and a discussion of deviant cases, which provide a clearer connection to the data and enrich the analysis.

Reduction in perceived stress

This theme, reported by 10 of the 11 participants, reflects the intervention's immediate impact on emotional regulation and stress reactivity. Parents and teachers observed that students developed a calmer demeanor and were better equipped to manage academic and social pressures. These improvements appear to be linked to students' increased use of mindful breathing and body awareness exercises during stressful situations, which enabled them to pause and respond rather than react impulsively.

Parent 1 said, "Since starting MBCT, my child seems calmer and more relaxed. They don't get as worked up about things anymore." Teacher 2 also emphasized, "Students who have been practicing mindfulness techniques are better able to handle stressful situations. They seem more composed and less reactive."

Nevertheless, not all students showed similar progress. Teacher 4 noted that "two students still avoid class participation under stress," suggesting that MBCT may require adaptation for those with more entrenched anxiety patterns or social withdrawal tendencies. This observation aligns with previous research indicating that mindfulness alone may be insufficient for students with severe avoidance behaviors unless combined with additional cognitive-behavioral support. *Interpreting this variation, students with high baseline avoidance may need graded exposure or brief skills coaching to translate mindful awareness into approach behavior; in our sample, such cases were also those with smaller PSS/PCS gains, indicating partial convergence between qualitative accounts and quantitative change.*

Improved psychological well-being

Nine participants reported that students demonstrated improved mood, increased emotional resilience, and heightened self-awareness. Parents described their children as "happier" and "more optimistic," while teachers noted greater emotional insight and articulation of feelings. These outcomes appear to stem from MBCT's emphasis on observing thoughts and emotions without judgment, which helped students reframe negative experiences.

Parent 3 stated, "My child's mood has improved noticeably since starting MBCT. They seem happier and more optimistic about the future." Teacher 1 observed, "Students can now explain what they feel—some even say, 'I'm overwhelmed,' which they never said before."

Still, Parent 2 noted, "Although my child is more self-aware, I wouldn't say they're emotionally resilient yet—it's a work in progress." This variation highlights that while MBCT fosters self-awareness early on, emotional resilience might require longer-term or reinforced practice to solidify. *Consistent with the study's mechanism-of-change lens, early gains in metacognitive awareness (decentering, labeling) likely precede durable affective resilience; this sequencing helps explain why some students showed strong mood reports before significant academic or social shifts.*

Enhanced social relationships

This theme, evident in 8 interviews, captures how MBCT facilitated empathy and improved peer interactions. Teachers reported that students were more attentive listeners and demonstrated greater patience and understanding during group activities. Parents similarly observed that their children were more willing to share experiences and engage in conversations, which improved family communication dynamics.

Parent 5 shared, "I've noticed my child becoming more open and communicative since starting MBCT. They listen more attentively and show genuine concern for their friends." Teacher 3 noted, "Students who practiced mindfulness exercises have become more empathetic and supportive of classmates."

Yet, Teacher 5 pointed out, "Some students remain socially reserved despite the intervention," indicating that while MBCT may enhance social-emotional awareness, individual personality factors, such as introversion or past social experiences, may moderate its impact. This suggests a need for complementary social skills training for students who remain reluctant to engage with peers. *Accordingly, we interpret social gains as contingent on trait introversion and prior peer history; where reticence persisted, the corresponding students typically showed modest improvements in PSS/PCS, again aligning theme-level narratives with measured outcomes.*

Improved academic engagement and performance

Seven participants mentioned that MBCT contributed to enhanced focus, attention, and persistence in academic tasks. Teachers reported noticeable improvements in students' ability to remain on task and engage actively in classroom discussions. These outcomes can be attributed to mindfulness techniques that promote attentional control and reduce cognitive interference caused by stress or negative emotions.

Parent 4 commented, "My child's grades have improved since they started MBCT. They're able to concentrate better and stay focused during study sessions." Teacher 1 added, "Students who have been practicing mindfulness show more enthusiasm for learning and participate more actively."

However, some parents felt these gains were gradual. Parent 6 remarked, "The academic improvements are there, but they're slower compared to changes in mood or behavior." This observation aligns with research indicating that cognitive and academic benefits from mindfulness typically emerge later than emotional outcomes, as they require sustained cognitive retraining. *This lag is consistent with a cascading-change account: stabilization of affect and stress first reduces interference (reflected in PSS/PCS), with study habits and performance consolidating more slowly under classroom demands.*

Strengthened stress management skills

Nine participants described how students became more adept at identifying stress triggers and employing mindful strategies to manage their stress. Teachers highlighted improvements in problem-solving abilities, while parents noted that their children paused and reflected before reacting to stressful events. These skills suggest that MBCT's emphasis on meta-awareness and non-reactivity effectively translated into adaptive coping.

Parent 2 shared, "My child has become more aware of their stress triggers and knows how to manage them better now." Teacher 4 observed, "Students demonstrate better problem-solving skills and are more resourceful in dealing with challenges."

Yet, Teacher 2 noted challenges in habit formation: "Some students say they forget to use the techniques when stressed, which shows that reinforcement might still be needed." This highlights the potential value of booster sessions or mobile-based mindfulness reminders in helping students maintain consistent practice. *We interpret these lapses as state-dependent retrieval failures; low-burden prompts (teacher cues, brief reminders) and booster sessions are plausible implementation levers to support skill access under peak arousal.*

Sustained behavioral change and long-term mindfulness integration

Six participants reported evidence of behavioral changes persisting beyond the intervention, including independent mindfulness practice and improved emotional balance. Parent 1 shared, "Even after the program ended, my child continues to practice mindfulness regularly. It's part of their daily routine." Teacher 3 commented, "Students report feeling more grounded and continue using techniques beyond the sessions."

However, not all students maintained these habits. Parent 6 admitted, "He used to meditate daily, but now only does it when reminded." This variability suggests that while MBCT can initiate long-term behavioral shifts, sustained reinforcement and support from parents or teachers are critical for habit consolidation. *We theorize that maintenance is a function of practice "dose," environmental scaffolds, and perceived usefulness. Students who reported routine supports (parent prompts, peer practice) also tended to show larger sustained quantitative gains, indicating cross-strand convergence.*

Discussion and conclusion

This study aimed to evaluate the effectiveness of mindfulness-based cognitive therapy (MBCT) in reducing perceived stress and improving mental health among adolescents with psychosomatic disorders. The findings suggest that MBCT had a significant positive impact on reducing perceived stress and improving mental health, aligning with previous studies supporting mindfulness interventions for psychological distress [1, 2]. *In applied terms, the observed mean reductions exceeded a conventional half-standard-deviation benchmark in this sample, which strengthens the case for clinical relevance; however, this threshold is heuristic and should be interpreted cautiously alongside confidence intervals.* However, the small sample size, quasi-experimental design, and lack of an active control group limit the generalizability of these findings. These factors, alongside the absence of blinding, may have contributed to potential placebo effects; therefore, future studies should incorporate larger, randomized controlled trials to isolate the specific effects of MBCT more effectively. *Specifically, expectancy and attention effects could inflate between-group contrasts when controls receive less contact; detection bias is also plausible when assessors are not fully blinded. These risks may bias estimates upward, implying that our effects represent an upper bound under these design constraints.*

The primary outcome of reduced perceived stress found in this study corroborates findings by Porcelli et al. [3], who demonstrated that mindfulness interventions can help alleviate stress in adolescents with psychosomatic disorders. Adolescents with such disorders often face heightened vulnerability to stress due to the

chronic interaction between physical symptoms and negative cognitive patterns [4]. MBCT appears to disrupt this cycle by promoting metacognitive awareness, enabling adolescents to observe their thoughts, emotions, and bodily sensations as transient events rather than fixed realities. Although these results are promising, the absence of an active control group and the lack of blinding warrant caution in interpreting the efficacy of MBCT, as placebo effects could not be ruled out. This mindfulness approach helps patients disengage from maladaptive thinking patterns, reducing the intensity of stress responses. Fundamentally, mindfulness fosters acceptance and a non-judgmental awareness of thoughts, which promotes cognitive, metacognitive, and behavioral strategies that focus attention on the present moment rather than on worries or future stressors [5]. *Future trials should test theorized mechanisms (e.g., changes in decentering, rumination, and resilience) as mediators of PSS/PCS change and examine moderators such as baseline avoidance, trait introversion, home-practice dose, and teacher fidelity to clarify for whom and under what conditions MBCT works best.*

The findings of this study are also consistent with recent evidence supporting the effectiveness of mindfulness-based interventions in reducing psychological distress and improving emotion regulation. For example, Feng et al. [67] demonstrated that mindfulness significantly reduced parenting stress among parents of children with autism, with resilience and psychological flexibility mediating this effect. Similarly, Zhang et al. [68] showed that mindfulness helped Chinese university students manage academic stress and improve emotional intelligence. A meta-analysis by Ng et al. [69] confirmed that mindfulness-based interventions are highly effective in reducing psychological symptoms in high school adolescents. Additionally, Lan et al. [70] found that mindfulness combined with life-skills training helped improve emotion regulation and reduce anxiety symptoms among Chinese migrant children. This body of research reinforces the notion that MBCT provides a robust framework for addressing both emotional and psychosomatic challenges in adolescents. However, it would be beneficial to incorporate cultural context into future studies to assess whether these interventions are equally effective in different cultural settings, particularly in how Chinese adolescents respond to such interventions. *In particular, collectivist norms, academic intensity, family-school coordination, and stigma surrounding help-seeking may shape both uptake and outcomes. Process evaluations that document cultural tailoring (such as language, metaphors, and parental involvement) and measure acceptability/appropriateness will improve interpretability and transferability.*

These therapeutic features suggest that MBCT encourages individuals to reconsider and reinterpret stressful experiences, reducing cognitive distortions such as catastrophizing and overgeneralization [10]. Moreover, MBCT's focus on being present in the moment helps individuals shift attention away from stress-inducing thoughts toward sensory experiences—such as bodily sensations, sounds, or mental images—thus alleviating the cognitive load caused by stress [5]. While these mechanisms are promising, further research is needed to explore how these cognitive and emotional changes evolve, particularly beyond the short-term follow-up period in this study. *Longer follow-ups (e.g., 6–12 months) with maintenance/booster supports and repeated mediator assessments would clarify durability and temporal sequencing of change.*

The second significant finding concerns the improvement in mental health outcomes for students with psychosomatic disorders. This aligns with findings from Lan et al. [9], Petrenko et al. [2], and Martin et al. [11], who emphasized MBCT's ability to enhance emotional resilience and reduce psychosomatic symptoms. By equipping individuals with cognitive and behavioral strategies, MBCT helps students manage stress and anxiety, thereby improving their ability to function effectively in daily life. Mindfulness techniques are known to reduce emotional distress and alleviate physical symptoms associated with psychosomatic conditions [12]. Through mechanisms such as enhanced body awareness and cognitive reframing, MBCT reduces avoidance behaviors and encourages acceptance of symptoms as part of a stress-response system, thereby reducing their severity. While these findings are significant, the absence of a placebo or active control group, as well as the lack of blinding in this study, are substantial limitations that could affect the robustness of the conclusions. *Moreover, the relative stability of the control group could reflect valid invariance, regression to the mean, or minor expectancy effects from administrative contact; an attention-matched or education-based comparator would help disentangle specific from non-specific benefits.*

A key consideration in this study is the cultural context, as the sample consisted of Chinese adolescents. While the intervention's effectiveness in reducing stress and improving mental health is evident, the cultural context is not adequately explored in this study. Given the distinct cultural values and practices in China, it is essential to understand how these factors may influence both the effectiveness and acceptability of the intervention. Future research should investigate how cultural factors—such as collectivism, attitudes toward mental health, and social expectations—affect the success of mindfulness-based interventions in non-Western contexts. *We recommend prespecifying culturally relevant implementation*

outcomes (acceptability, feasibility, appropriateness), documenting adaptations using standardized frameworks, and conducting multi-site studies to test transportability across provinces and school types.

In conclusion, this study contributes valuable insights into the potential of MBCT to reduce stress and improve mental health among adolescents with psychosomatic symptoms. By integrating both quantitative outcomes and qualitative insights from parents and teachers, the research offers a comprehensive view of the impact of MBCT on adolescents' well-being. However, the study's limitations, such as the small sample size, the quasi-experimental design, and the lack of an active control group, suggest that these findings should be interpreted with caution. *Generalizability is bounded by setting (three public schools in DaZhou), age (14–16), and gender (female), and the design likely overestimates effects relative to fully blinded, attention-matched RCTs.* To strengthen the validity of future research, studies should utilize randomized controlled trials, include larger and more diverse samples, and incorporate extended follow-up periods to assess the durability of the intervention's effects. Additionally, psychological mechanisms underlying the efficacy of MBCT should be explored in more detail, focusing on the cognitive and emotional processes involved in these interventions. *Specifically, we advocate for multi-arm trials comparing MBCT with an attention-matched health education control, preregistered mediator analyses (decentering, rumination, resilience), and moderator analyses (baseline avoidance, practice dose, instructor fidelity), as well as cost-effectiveness assessments, to inform school policy.* Finally, the cultural context should be carefully considered, particularly when implementing MBCT in non-Western populations, to ensure that the intervention is both practical and culturally appropriate. *Adopting a cautious interpretive stance until such trials are complete will prevent overgeneralization and guide responsible scaling.*

Practical implications

Given the demonstrated benefits of mindfulness-based cognitive therapy (MBCT) in reducing perceived stress and improving mental health outcomes, integrating MBCT into school-based psychological services appears to be a promising avenue for enhancing adolescent well-being. However, the real-world implementation of MBCT in educational settings requires addressing several systemic and logistical challenges. One significant barrier is the limited availability of trained counselors in schools. Many schools already face a shortage of mental health professionals, and adding MBCT as a standard intervention could stretch existing resources. To overcome this, schools could explore partnerships with external mental health professionals or community

organizations to provide additional support, especially in underserved areas. *As a concrete pathway, districts can adopt a “train-the-trainer” model: (i) certify 2–3 lead counselors per school (24–32 h of MBCT-specific training plus supervised practice), (ii) cascade 6–8 h of basic mindfulness classroom skills to homeroom teachers, and (iii) schedule monthly supervision huddles to maintain fidelity and address implementation drift. Staffing plans should specify caseload ratios (e.g., ≤ 12 students per facilitator per cycle), release time for teachers, and coverage for sessions to minimize academic disruption.*

Time constraints within academic schedules present another challenge. School schedules are already packed with core educational content, and finding time for MBCT sessions could be difficult. To address this, MBCT could be integrated into existing programs such as social-emotional learning (SEL) curricula or offered as part of extracurricular activities. Another option could be offering short, weekly mindfulness sessions during breaks or advisory periods, which would allow students to engage in MBCT without disrupting their academic studies. The development of flexible, brief interventions that require minimal time commitment from students could increase feasibility and adoption in schools. *Operationally, schools can pilot two delivery formats and compare feasibility: (a) the full protocol (8 × 75 min after school) versus (b) a micro-dose format (10 × 20 min embedded in advisory/SEL blocks), with the latter supplemented by 5-minute “transition practices” at the start of classes. A scheduling matrix that avoids peak exam periods and allows make-up sessions will reduce attrition.*

Additionally, to effectively implement MBCT, there is a need for culturally adapted training materials and resources. As highlighted by recent studies, cultural factors play an essential role in the acceptability and effectiveness of psychological interventions, particularly in diverse educational settings [1]. In the context of non-Western populations, like Chinese adolescents, mindfulness interventions may require adjustments to better align with local cultural values and beliefs about mental health. Developing culturally relevant training programs for counselors and teachers would ensure that MBCT resonates with students and meets their unique psychological and social needs. For example, incorporating local beliefs about stress, emotional regulation, and coping could enhance the intervention's relevance and impact. *Illustratively, scripts can use culturally resonant metaphors (e.g., 调息/tiao-xi for “regulating breath,” 修身/xiūshēn for self-cultivation), pair practices with collectivist values (classroom harmony, filial respect), and include family-friendly home activities that fit exam-heavy routines (e.g., 3-minute evening body scan before homework). Teacher materials should provide bilingual key terms, examples from Chinese classroom life (e.g.,*

recitation pressure), and brief guidance for stigma-sensitive discussions about mental health. Where possible, parent orientation slides and WeChat mini-modules can normalize practice and invite supportive routines at home.

A feasible next step for the broader implementation of MBCT in schools could be the development of pilot programs and workshops for teachers and counselors. These pilot programs could serve as testing grounds for evaluating the scalability of MBCT in different educational settings. By running small-scale trials, schools can assess the practical challenges of implementation, gather feedback from participants, and fine-tune the delivery of MBCT to maximize its effectiveness and accessibility. Teachers and counselors could receive specialized training on how to facilitate mindfulness exercises and incorporate MBCT principles into their daily interactions with students. These programs could also help familiarize school staff with mindfulness-based strategies, which could be beneficial in fostering a supportive school climate that prioritizes mental health. *We recommend a 6–9-month pilot framework with clear milestones: Month 1—staff training and fidelity rehearsal; Months 2–4—first student cohort (randomized to MBCT vs. attention-matched health education), Months 5–6—second cohort; Months 7–9—data synthesis and scale-up planning. Monitor both effectiveness (PSS/PCS pre/post/follow-up) and implementation outcomes (acceptability, feasibility, appropriateness, fidelity), and include teacher-rated classroom behavior as a pragmatic metric. Supervision checklists, session observation rubrics, and brief student practice logs will help maintain fidelity while identifying training gaps.*

To further support students' independent practice and enhance the sustainability of intervention effects, schools could incorporate digital mindfulness tools, such as guided audio exercises, mobile applications, or online platforms. These tools could provide students with additional resources to practice mindfulness outside of scheduled sessions, reinforcing the skills learned during MBCT and encouraging self-regulation. Digital tools provide an accessible and scalable solution to support the ongoing practice of mindfulness, particularly in schools where time and staffing resources are limited. Mobile apps with mindfulness exercises could also help engage students who might be reluctant to participate in traditional face-to-face sessions, providing them with a more private and flexible means of integrating mindfulness into their daily lives. Moreover, these tools can offer real-time tracking and feedback, providing valuable insights into students' progress and enabling educators to personalize interventions as needed. *In the Chinese school context, low-friction options include QR-coded audio libraries hosted on the school's LMS and WeChat mini-programs with 3–5 min practices, offline access for bandwidth-limited*

students, and opt-in push reminders before study periods. Aggregated, de-identified usage analytics (practice “dose”) can be linked to outcomes to inform targeted supports, while privacy protocols should be aligned with local regulations and school policies. Given the qualitative evidence for the role of families, brief parent micro-modules (e.g., “how to prompt without pressuring”) and at-home dyadic practices can extend benefits beyond the classroom and improve habit maintenance.

Supplementary Information

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Supplementary Material 1

Author contributions

YunNan Zhao designed the study. YunNan Zhao collected the data. YunNan Zhao analyzed and interpreted the data. YunNan Zhao drafted the manuscript. YunNan Zhao proofread the paper. YunNan Zhao agreed to be accountable and verified the submitted version.

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Data availability

Data is provided within the manuscript.

Declarations

Ethics approval and consent to participate

The Institutional Review Board of Sichuan University of Arts and Science approved this study and issued a letter, confirming that the survey has no negative effects on the participants. All methods were carried out in accordance with relevant guidelines and regulations. Informed consent was obtained from all subjects, and for participants under the age of 16, it was obtained from their parents and/or legal guardian(s).

Consent for publication

Not applicable **because no individual-level identifiable data were reported.**

Competing interests

The authors declare no competing interests.

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